9 Illustrate the concept of inter-process communication using shared memory with a C program

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/ipc.h>
#include <sys/shm.h>

#define SHM_SIZE 1024   // Size of shared memory

int main()
{
    key_t key = ftok("shmfile", 65);

    // Create shared memory
    int shmid = shmget(key, SHM_SIZE, IPC_CREAT | 0666);
    if (shmid == -1)
    {
        perror("shmget");
        exit(EXIT_FAILURE);
    }

    // Attach memory
    char *shm_ptr = (char *)shmat(shmid, NULL, 0);
    if (shm_ptr == (char *)(-1))
```

```
    {
        perror("shmat");
        exit(EXIT_FAILURE);
    }

    // Write data
    strcpy(shm_ptr, "Hello, shared memory!");

    // ✅ Print BEFORE detaching
    printf("Data written to shared memory: %s\n", shm_ptr);

    // Detach memory
    if (shmdt(shm_ptr) == -1)
    {
        perror("shmdt");
        exit(EXIT_FAILURE);
    }

    // Remove shared memory
    if (shmctl(shmid, IPC_RMID, NULL) == -1)
    {
        perror("shmctl");
        exit(EXIT_FAILURE);
    }

    return 0;
}
```

OUTPUT:

```
Output

Data written: Hello, Shared Memory!
```